

Image Derivatives and Edge Detection

1. Do the following
 - (a) Calculate the gradient in x and gradient in y using the `SOBEL` build-in function in OpenCV.
 - (b) Calculate the gradient magnitude and threshold.
2. Calculate the Canny edge detector using the build-in function `Canny()`.
3. Do the following
 - (a) Do calculate the Laplacian (Double degree derivative).
 - (b) Find the Zero crossing using OpenCV.
 - (c) Find the Slope of Line using OpenCV.

Note: The submitted assignment should consist of a screen shot on a page showing the code and output result.